2. What is AWS IAM?

Answer

IAM (Identity and Access Management) is used to securely manage access to AWS resources.

Components:

- User
- Group
- Role
- Policy

---

3. Difference between IAM User and Role?

User	Role
Permanent credentials	Temporary credentials
Human login	AWS Service
Password	No Password

---

4. What is Assume Role?

Allows temporary access to AWS resources using STS.

Example:

EC2 → S3

Lambda → DynamoDB

Cross-account access

---

5. Explain Least Privilege.

Give users only the permissions required to perform their job.

Example:

Developer should not have AdministratorAccess.

---

6. Difference between Security Group and NACL?

Security Group	NACL
Stateful	Stateless
Instance Level	Subnet Level
Allow Only	Allow & Deny

---

7. Explain VPC.

A Virtual Private Cloud is an isolated network inside AWS.

Components:

- CIDR
- Public Subnet
- Private Subnet
- Route Table
- Internet Gateway
- NAT Gateway

- Security Group
- NACL

---

8. Public vs Private Subnet?

Public

- Internet Gateway
- Public IP

Private

- NAT Gateway
- No Public IP

---

9. Explain Auto Scaling.

Automatically increases or decreases EC2 instances based on CPU, memory, or custom metrics.

---

10. Difference between ALB and NLB?

ALB	NLB
Layer 7	Layer 4
HTTP	TCP
Path Routing	High Performance

---

11. What is S3?

Object Storage.

Uses:

- Backup
- Static Website
- Log Storage
- Terraform Backend

---

12. Difference between S3, EBS and FSx?

S3	EBS	FSx
Object	Block	File
Unlimited	EC2 Disk	Shared Storage

Terraform

13. Explain Terraform.

Infrastructure as Code tool.

Commands

terraform init

terraform plan

terraform apply

terraform destroy

---

14. What is Terraform State?

Stores resource mapping.

Best Practice

S3 + DynamoDB

---

15. Why Remote Backend?

- Team Collaboration
- Locking
- Backup
- Versioning

---

16. Terraform Module?

Reusable code.

Example

modules/

vpc/

eks/

iam/

---

17. CloudFormation vs Terraform?

Terraform

Multi Cloud

CloudFormation

AWS Only

CI/CD

18. Explain CI/CD.

CI

Build

Test

Scan

CD

Deploy

Monitor

---

19. GitHub Actions Workflow?

Developer

↓

GitHub

↓

Workflow

↓

Docker Build

↓

ECR

↓

Deploy

---

20. Jenkins Pipeline Stages?

Checkout

Build

Test

Docker Build

Push

Deploy

---

21. What is Ansible?

Configuration Management Tool.

Agentless.

Uses SSH.

Playbooks in YAML.

---

22. Why JFrog Artifactory?

Stores

- JAR
- Docker Images
- npm Packages
- Maven Packages

Git

23. GitFlow?

Branches

Main

Develop

Feature

Release

Hotfix

---

24. Trunk Based Development?

Small commits directly into main branch.

---

25. Git Rebase vs Merge?

Merge

Creates Merge Commit

Rebase

Linear History

Docker

26. Docker Lifecycle?

Dockerfile

↓

Build

↓

Image

↓

Container

---

27. Docker Image vs Container?

Image

Template

Container

Running Image

---

28. Multi-stage Build?

Reduces image size.

Kubernetes

29. What is Kubernetes?

Container orchestration platform.

---

30. Pod vs Deployment?

Pod

Single Instance

Deployment

Multiple Pods

Rolling Update

---

31. ReplicaSet?

Maintains desired pod count.

---

32. Deployment Strategy?

Rolling Update

Recreate

Blue Green

Canary

---

33. ConfigMap vs Secret?

ConfigMap

Non Sensitive

Secret

Sensitive Data

---

34. Namespace?

Logical Isolation.

---

35. Ingress?

HTTP Routing.

SSL.

Load Balancing.

---

36. Service Types?

ClusterIP

NodePort

LoadBalancer

ExternalName

---

37. PV vs PVC?

PV

Storage

PVC

Storage Request

---

38. Readiness vs Liveness?

Readiness

Ready for Traffic

Liveness

Restart Container

Python

39. Python script to list EC2

```
import boto3


ec2 = boto3.client("ec2")


response = ec2.describe_instances()


for reservation in response["Reservations"]:
    for instance in reservation["Instances"]:
        print(instance["InstanceId"])
```

PowerShell

40. Restart IIS

iisreset

Restart Service

Restart-Service Spooler

Windows

41. How do you troubleshoot Windows Server?

Check

Services

Event Viewer

Disk Space

CPU

Memory

Network

Firewall

---

42. RDP Not Working?

Check

3389

Firewall

Security Group

Service

Scenario Questions

43. EC2 not reachable?

Check

Security Group

NACL

Route Table

IGW

Public IP

SSH

---

44. Terraform Apply Failed?

Check

Logs

Credentials

Permissions

Syntax

State

---

45. GitHub Action Failed?

Check

Workflow

Secrets

Logs

Docker Build

AWS Credentials

---

46. Docker Image Pull Failed?

Check

Registry

Image Tag

Authentication

Network

---

47. Pod CrashLoopBackOff?

kubectl logs

kubectl describe pod

kubectl get events

---

48. Application Slow?

Check

CPU

Memory

Database

Logs

Grafana

---

49. Production Deployment Failed?

Rollback

Check Logs

Fix

Deploy Again

Verify Health

51. Explain your end-to-end CI/CD pipeline.

Answer

In my current project, the CI/CD pipeline works as follows:

1. Developer pushes code to GitHub.
2. GitHub Actions triggers automatically.
3. Source code is checked out.
4. Unit tests are executed.
5. Docker image is built.
6. Trivy scans the image for vulnerabilities.
7. Docker image is pushed to Amazon ECR.
8. Helm values file is updated with the new image tag.
9. ArgoCD detects the Git change and deploys the application to Amazon EKS.
10. Prometheus and Grafana monitor the deployment.

---

52. What is Blue-Green Deployment?

Answer

Blue-Green Deployment uses two identical environments.

- Blue = Current production
- Green = New version

Traffic is switched from Blue to Green after successful validation.

Advantages:

- Zero downtime
- Easy rollback
- Safer releases

---

53. What is Canary Deployment?

Answer

A new version is deployed to a small percentage of users first (e.g., 10%). If it performs well, traffic is gradually increased until all users are served by the new version.

---

54. Explain Rolling Update.

Answer

Pods are updated one by one while keeping the application available.

Benefits:

- Zero downtime
- Automatic rollback if configured
- No service interruption

---

55. What is HPA?

Answer

Horizontal Pod Autoscaler automatically scales pods based on metrics such as CPU or memory utilization.

Example:

kubectl autoscale deployment app --cpu-percent=70 --min=2 --max=10

---

56. Difference between HPA and Cluster Autoscaler?

HPA	Cluster Autoscaler
Scales Pods	Scales Nodes
CPU/Memory based	Pending Pods based
Application level	Cluster level

---

57. What is an Init Container?

Answer

An Init Container runs before the main application container.

Use cases:

- Database migration
- Waiting for a dependent service
- Downloading configuration files

---

58. Explain Kubernetes Scheduler.

Answer

The Scheduler decides which node should run a pod based on:

- Available CPU and memory
- Taints and tolerations
- Node affinity
- Resource requests

---

59. What happens when a Node fails?

Answer

- Kubernetes marks the node as **NotReady**.
- Pods become unavailable.
- ReplicaSet/Deployment recreates pods on healthy nodes.
- Traffic is redirected automatically.

---

60. Explain Taints and Tolerations.

Answer

Taints prevent pods from being scheduled on certain nodes.

Tolerations allow specific pods to run on tainted nodes.

Example: Reserve GPU nodes for machine learning workloads.

---

61. Explain Node Affinity.

Answer

Node Affinity ensures pods are scheduled only on nodes with matching labels.

Example:

```
nodeSelector:
  environment: production
```

---

62. Difference between StatefulSet and Deployment?

Deployment	StatefulSet
Stateless applications	Stateful applications
Random pod names	Stable pod names
No persistent identity	Persistent identity
Frontend/API	Database

---

63. How do you debug ImagePullBackOff?

Answer

1. Verify the image name and tag.
2. Check if the image exists in the registry.
3. Validate image pull secrets.
4. Confirm network connectivity.
5. Review pod events:

```
kubectl describe pod
```

---

64. How do you debug OOMKilled?

Answer

- Check memory usage.
- Increase memory limits if required.
- Analyze application memory leaks.
- Review `kubectl describe pod` events.

---

65. Explain Docker Networking.

Answer

Docker supports:

- Bridge
- Host
- Overlay
- None
- Macvlan

The default network is **Bridge**.

---

66. Difference between COPY and ADD?

COPY	ADD
Copies files	Can also extract archives and download URLs
Preferred	Use only when additional features are required

---

67. What is Docker Volume?

Answer

Volumes store persistent data outside the container lifecycle.

Benefits:

- Data persistence
- Easier backups
- Data sharing between containers

---

68. Explain Docker Registry.

Answer

A Docker Registry stores container images.

Examples:

- Amazon ECR
- Docker Hub
- JFrog Artifactory
- Harbor

---

69. Explain Route53.

Answer

Amazon Route53 is AWS's managed DNS service.

Features:

- Domain registration
- DNS routing
- Health checks
- Failover routing
- Weighted routing

---

70. Difference between ALB and Classic Load Balancer?

ALB	Classic LB
Layer 7	Layer 4 & 7
Host/path routing	Basic routing
Modern applications	Legacy applications

---

71. What is NAT Gateway?

Answer

A NAT Gateway enables instances in private subnets to access the internet without exposing them to inbound internet traffic.

---

72. Explain Internet Gateway.

Answer

An Internet Gateway allows resources in public subnets to communicate with the internet.

---

73. Difference between Bastion Host and NAT Gateway?

Bastion Host	NAT Gateway
SSH/RDP access	Internet access for private instances
Manual login	No login capability

---

74. Explain AWS Security Best Practices.

Answer

- Enable MFA.
- Follow least-privilege access.
- Rotate credentials.
- Encrypt data at rest and in transit.
- Enable CloudTrail.
- Enable AWS Config.
- Use IAM Roles instead of access keys.

---

75. What is CloudTrail?

Answer

CloudTrail records all AWS API calls for auditing and compliance.

---

76. What is CloudWatch?

Answer

CloudWatch collects:

- Metrics
- Logs
- Events
- Alarms
- Dashboards

---

77. Difference between CloudWatch and CloudTrail?

CloudWatch	CloudTrail
Monitoring	Auditing
Metrics and Logs	API activity

---

78. Explain Ansible Inventory.

Answer

An inventory lists managed hosts.

Example:

```
[web]
10.0.1.10
10.0.1.11

[db]
10.0.2.10
```

---

79. Explain Ansible Playbook.

Answer

A Playbook is a YAML file that defines automation tasks.

Example:

```
- hosts: web
  become: yes
  tasks:
    - name: Install nginx
      apt:
        name: nginx
        state: present
```

---

80. Explain Git Rebase.

Answer

Git Rebase moves commits onto a new base, creating a cleaner, linear commit history.

---

81. What is Git Cherry-pick?

Answer

`git cherry-pick` copies a specific commit from one branch to another without merging the entire branch.

---

82. Explain Merge Conflict.

Answer

A merge conflict occurs when Git cannot automatically combine changes because the same lines were modified in different branches. Resolve the conflict manually, test the code, and then complete the merge.

---

83. Explain JFrog Artifactory.

Answer

JFrog Artifactory is an artifact repository that stores:

- Docker images
- Maven packages
- npm packages
- Python packages
- Generic binaries

---

84. How do you secure CI/CD pipelines?

Answer

- Store secrets securely.
- Scan code and images.
- Use branch protection.
- Enable RBAC.
- Require code reviews.
- Implement approval gates.
- Sign artifacts when appropriate.

---

85. Explain Infrastructure as Code.

Answer

Infrastructure as Code (IaC) means provisioning and managing infrastructure through code instead of manual processes. This provides consistency, version control, repeatability, and automation.

---

86. Explain Idempotency.

Answer

An idempotent operation produces the same result regardless of how many times it is executed.

Example: Running the same Ansible playbook multiple times will not reinstall an already-installed package.

---

87. Production server is down. What will you do?

Answer

1. Check monitoring alerts.
2. Verify EC2/EKS health.
3. Review application logs.
4. Identify the root cause.
5. Roll back if needed.
6. Restore service.
7. Perform root cause analysis (RCA).
8. Document preventive actions.

---

88. Database is not reachable. How do you troubleshoot?

Answer

- Check database service status.
- Verify security groups and NACLs.
- Confirm DNS resolution.
- Test network connectivity.
- Review database logs.
- Validate credentials.
- Check resource utilization.

---

89. CPU utilization is 100%. What will you do?

Answer

- Identify the process consuming CPU.
- Check application logs.
- Scale the application (HPA/Auto Scaling).
- Tune resource requests and limits.
- Investigate inefficient code or queries.

90. How do you handle production deployments?

Answer

- Take backups.
- Verify deployment artifacts.
- Deploy during a maintenance window if needed.
- Perform smoke tests.
- Monitor logs and metrics.
- Roll back immediately if issues occur.

---

91. What happens when you launch an EC2 instance?

Answer:

1. AMI is selected.
2. VPC and subnet are chosen.
3. ENI (Elastic Network Interface) is attached.
4. Security Group is attached.
5. EBS volume is attached.
6. IAM Role is attached (optional).
7. Instance boots using the hypervisor.
8. User Data script executes (if configured).

---

92. What is an AMI?

An **Amazon Machine Image (AMI)** is a template containing:

- Operating System
- Application software
- Configuration
- Permissions

It is used to launch EC2 instances consistently.

---

93. Explain the EC2 boot process.

BIOS/UEFI → Hypervisor → Kernel → Operating System → Cloud-Init/User Data → Application Startup.

---

94. Difference between EBS and Instance Store?

EBS	Instance Store
Persistent	Temporary
Network-attached	Physically attached
Snapshots supported	No snapshots
Recommended for production	Suitable for temporary/cache data

---

95. What is an Elastic IP?

A static public IPv4 address that can be remapped to another EC2 instance.

---

96. What is CloudFormation Drift Detection?

It identifies differences between the deployed infrastructure and the CloudFormation template.

---

97. What is Terraform Drift?

Terraform detects infrastructure changes made outside Terraform by comparing the current infrastructure with the Terraform state during terraform plan.

---

98. What are Terraform Workspaces?

Workspaces allow managing multiple environments (e.g., dev, test, prod) using the same Terraform configuration while maintaining separate state files.

---

99. Explain Terraform Lifecycle Meta-Arguments.

Examples include:

```
lifecycle {
  create_before_destroy = true
  prevent_destroy       = true
  ignore_changes        = [tags]
}
```

---

100. What are Terraform Data Sources?

Data sources allow Terraform to read information about existing infrastructure without creating it.

Example:

```
data "aws_vpc" "default" {}
```

101. Explain the Kubernetes Control Plane.

Components:

- API Server
- Scheduler
- Controller Manager
- etcd

The Control Plane manages the cluster state.

---

102. What is etcd?

A distributed key-value database used by Kubernetes to store cluster configuration and state.

---

103. What is kubelet?

An agent running on every worker node that communicates with the API Server and ensures containers are running as expected.

---

104. What is kube-proxy?

It manages network rules and enables communication between Services and Pods.

---

105. What is CoreDNS?

CoreDNS provides internal DNS resolution for Kubernetes services.

Example:

backend.default.svc.cluster.local

---

106. What is a DaemonSet?

A DaemonSet ensures one pod runs on every node.

Common examples:

- Fluent Bit
- Prometheus Node Exporter
- Security agents

---

107. What is a Job?

Runs a task until it completes successfully.

Example:

- Database migration
- Data import

---

108. What is a CronJob?

Runs Jobs on a schedule.

Example:

Every day at 2 AM

---

109. What is a ServiceAccount?

A Kubernetes identity used by Pods to interact with the Kubernetes API securely.

---

110. What is a Network Policy?

A Network Policy controls which Pods can communicate with other Pods or external services.

---

111. How do you reduce Docker image size?

- Use Alpine-based images.
- Use multi-stage builds.
- Remove unnecessary packages.

- Combine RUN commands.
- Clean package caches.

---

112. What is the difference between CMD and ENTRYPOINT?

CMD	ENTRYPOINT
Default command	Main executable
Can be overridden	Typically fixed

---

113. What happens during docker build?

1. Docker reads the Dockerfile.
2. Executes each instruction in order.
3. Creates image layers.
4. Caches reusable layers.
5. Produces the final image.

---

114. Explain Docker Layer Caching.

Docker reuses unchanged layers from previous builds to speed up image creation.

---

115. What is Docker Compose?

Docker Compose manages multi-container applications using a docker-compose.yml file.

---

116. Difference between Git Fetch and Git Pull?

Fetch	Pull

Downloads changes only	Downloads and merges changes

---

117. What is Git Reset?

git reset moves the branch pointer and can unstage or discard changes depending on the mode (--soft, --mixed, --hard).

---

118. What is Git Revert?

Creates a new commit that reverses the changes introduced by a previous commit without rewriting history.

---

119. Declarative vs Scripted Pipeline?

Declarative	Scripted
Simpler syntax	More flexible
Easier maintenance	More control

---

120. What are Jenkins Agents?

Agents execute build jobs assigned by the Jenkins Controller.

---

121. What are GitHub Runners?

Runners are machines that execute GitHub Actions workflows. They can be GitHub-hosted or self-hosted.

---

122. What are GitHub Secrets?

Encrypted values used to store sensitive information such as AWS credentials, API keys, and tokens.

---

123. What is Idempotency in Ansible?

Running the same playbook multiple times results in the same desired state without making unnecessary changes.

---

124. What are Ansible Roles?

Roles organize playbooks into reusable components with directories for tasks, handlers, templates, files, and variables.

---

125. Why use Boto3?

Boto3 is the AWS SDK for Python, used to automate AWS services such as EC2, S3, IAM, and Lambda.

---

126. How do you handle exceptions in Python?

```
try:
    print("Process")
except Exception as e:
    print(e)
```

Monitoring

127. Explain Prometheus Architecture.

Prometheus periodically scrapes metrics from configured targets, stores them in a time-series database, and allows querying with PromQL. Grafana visualizes these metrics.

---

128. What is Grafana?

Grafana is a visualization platform used to create dashboards for metrics, logs, and alerts.

---

129. Difference between Prometheus and CloudWatch?

Prometheus	CloudWatch
Open source	AWS managed
Pull-based	Native AWS monitoring
PromQL	CloudWatch Metrics

---

130. How do you secure Kubernetes?

- RBAC
- Network Policies
- Pod Security Standards
- Secret management
- Image scanning (Trivy)
- Regular upgrades
- Audit logging

---

131. How do you secure Docker images?

- Use trusted base images.
- Scan images with Trivy.
- Remove unnecessary packages.
- Avoid running containers as root.
- Sign images where appropriate.

132. Explain Principle of Least Privilege.

Grant only the minimum permissions required to perform a task.

Scenario Questions

133. A Pod is Pending. How do you troubleshoot?

Check:

- Available nodes
- Resource requests
- Node taints
- PVC binding
- Scheduler events

Command:

kubectl describe pod <pod-name>

---

134. Nodes are NotReady. What will you check?

- Node health
- kubelet status
- Network connectivity
- Disk space
- CPU and memory
- Cloud provider status

---

135. Deployment stuck in Progressing. What will you do?

- Review rollout status.
- Check ReplicaSets.
- Review pod events and logs.

- Verify readiness probes.
- Roll back if necessary.

---

136. Jenkins pipeline suddenly fails. What is your approach?

1. Review pipeline logs.
2. Identify the failed stage.
3. Check credentials.
4. Verify external dependencies.
5. Test manually if needed.
6. Fix and rerun.

---

137. High latency after deployment. What will you do?

- Compare with previous release.
- Check application logs.
- Review CPU and memory.
- Analyze database performance.
- Roll back if required.

---

138. How do you perform Root Cause Analysis (RCA)?

1. Collect logs and metrics.
2. Identify the triggering event.
3. Determine the root cause.
4. Implement a fix.
5. Add preventive measures.
6. Document findings and lessons learned.

---

139. How do you handle a Sev-1 Production Incident?

- Acknowledge the incident immediately.
- Assemble the response team.

- Restore service as quickly as possible.
- Keep stakeholders updated.
- Collect evidence during the incident.
- Conduct a post-incident RCA and implement preventive actions.